

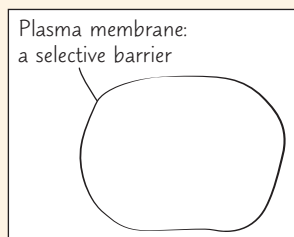
6 A Tour of the Cell

KEY CONCEPTS

- 6.1** Biologists use microscopes and biochemistry to study cells *p. 94*
- 6.2** Eukaryotic cells have internal membranes that compartmentalize their functions *p. 97*
- 6.3** The eukaryotic cell's genetic instructions are housed in the nucleus and carried out by the ribosomes *p. 102*
- 6.4** The endomembrane system regulates protein traffic and performs metabolic functions *p. 104*
- 6.5** Mitochondria and chloroplasts change energy from one form to another *p. 109*
- 6.6** The cytoskeleton is a network of fibers that organizes structures and activities in the cell *p. 112*
- 6.7** Extracellular components and connections between cells help coordinate cellular activities *p. 118*
- 6.8** A cell is greater than the sum of its parts *p. 121*

Study Tip

Draw animal and plant cells: Draw an outline of an animal cell and add structures, labels, and functions. Draw a plant cell labeled with structures unique to plant cells.



➔ Go to Mastering Biology

For Students (in eText and Study Area)

- Get Ready for Chapter 6
- Figure 6.7 Walkthrough: Geometric Relationships between Surface Area and Volume
- BioFlix® Animations: Tour of an Animal Cell and Tour of a Plant Cell

For Instructors to Assign (in Item Library)

- Tutorial: Tour of an Animal Cell: The Endomembrane System
- Tutorial: Tour of a Plant Cell: Structures and Functions



Figure 6.1 The cell is an organism's basic unit of structure and function. Many forms of life exist as single-celled organisms, such as the *Paramecium* shown here. Larger, more complex organisms, including plants and animals, are multicellular. In this chapter, we focus mainly on eukaryotic cells—cells with a nucleus.

How does the internal organization of eukaryotic cells allow them to perform the functions of life?

Internal membranes divide a cell, such as this plant cell, into compartments where specific chemical reactions occur.

Energy and matter transformations

A system of internal membranes synthesizes and modifies proteins, lipids, and carbohydrates.

Chloroplasts convert light energy to chemical energy.

Mitochondria break down molecules, generating ATP.

Interactions with the environment

The plasma membrane controls what goes into and out of the cell.

Plant cells have a protective cell wall.

Genetic information storage and transmission

DNA in the nucleus contains instructions for making proteins.

Ribosomes are the sites of protein synthesis.

